

Mohamed Tarek Torky

Software Engineer · Full-Stack · AI/ML

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 Egypt

Software Engineer with 2+ years of project and professional experience building and shipping production-ready full-stack systems. Specialized in React/Node.js architecture, REST API design, MongoDB performance optimization, and applied AI/ML integration. Proven track record of delivering measurable impact — from a 3,000-user student portal to a fine-tuned BERT model achieving 89% accuracy. Champion at the Microsoft APAC AI Accessibility Hackathon. AWS-certified across Cloud, Data Engineering, and Machine Learning. Strong DSA fundamentals, Agile delivery experience, and a consistent record of academic excellence (CGPA 3.82 / 4.00).

 PROFESSIONAL EXPERIENCE

Mirco Semiconductor Sdn Bhd

Software Engineer

Sep 2023 – Feb 2024

Johor, Malaysia

- Enhanced and maintained **3+ internal and client-facing web applications**, improving page load performance by **32%** through frontend optimization, code-splitting, and asset caching.
- Architected a **secure CV Generator system** in Laravel with multi-role authentication, server-side validation, and automated PDF generation, cutting manual HR processing time from ~4 hours/week to under 2 hours.
- Operated in **Agile Scrum sprints**; owned **20+ Jira tasks** across **3 sprint cycles**, maintaining zero missed deadlines across the engagement

Project-Based Professional Experience

Jan 2023 – Present

- Lead Full-Stack Engineer · ISS Egypt Gateway — Student Portal Platform** 
MERN Stack · MongoDB · Express · React · Node.js · 3,000+ active users
 - Architected and delivered a **full-stack student portal** from zero to production, serving **3,000+ active users**, supporting academic, social, and administrative workflows.
 - Designed and implemented **role-based access control (RBAC)** with **JWT authentication**, securing **5+ user tiers** and **30+ protected API routes**.
 - Optimized MongoDB aggregation pipelines and indexing strategy, reducing average query time by **60%** under concurrent load — enabling <100ms p95 API response time.
- Full-Stack Engineer · Nexus 4D — Gamified Learning Management System (LMS)** 
MERN Stack · React · Node.js · MongoDB · Cloudflare R2
 - Engineered a high-performance **drag-and-drop Course Editor** supporting **5+ media types**, reducing course creation time by **40** through optimistic UI updates and debounced autosave.
 - Built a real-time **Gamification Engine** (XP, Levels, Leaderboards) using MongoDB change streams, targeting a **35%** lift in daily active usage
 - Constructed an **Admin Analytics Dashboard** processing **10,000+ data points** to visualize revenue, user growth, and engagement KPIs.
 - Optimized backend performance, reducing **API response times to <100ms** through MongoDB indexing and aggregation pipelines.
- Full-Stack AI Engineer · E-GPT — AI Chatbot Platform** 
Node.js · MongoDB Atlas Vector Search · OpenAI GPT · Google Gemini
 - Developed a **bilingual (English/Arabic) RAG chatbot** platform using MongoDB Atlas **Vector Search** for semantic retrieval against a curated **100+ question** knowledge base.
 - Integrated **OpenAI ChatGPT** and **Google Gemini** with model-fallback logic, ensuring high availability and graceful degradation under API rate limits.
- ML Engineer · Airline Sentiment Intelligence Platform (NLP & BI)**
Python · PyTorch · Hugging Face Transformers · Power BI
 - Engineered an **end-to-end NLP pipeline** processing **41,000+** unstructured airline passenger reviews — from raw data ingestion through feature engineering to deployment-ready inference.
 - Fine-tuned a **BERT transformer model** (PyTorch, Hugging Face), achieving **89% accuracy** and a **0.88 F1-score** for negative sentiment detection.
 - Delivered an interactive **Power BI dashboard**, revealing that **60% of dissatisfied business travelers** cited baggage handling and operational delays as key pain points.

 EDUCATION

Master's in Data Science

Universiti Teknologi Malaysia (UTM)

- CGPA: 3.82 / 4.00** · Specialization in machine learning, NLP, and BI analytics

Bachelor of Computer Science (Honours) – Software Engineering

Universiti Teknologi Malaysia (UTM) 

- CGPA: 3.79 / 4.00** · Dean's List — 7 semesters

AWARDS & RECOGNITION

Champion — Microsoft APAC AI Accessibility Hackathon

Jun 2023

Microsoft · Asia-Pacific region

- Won 1st place out of regional teams for "A-EYE" — an AI-powered real-time auditory navigation system for visually impaired users, built with computer vision and speech synthesis within a 48-hour hackathon sprint.

Dean's List — 7 Consecutive Semesters

2020 – 2024

Universiti Teknologi Malaysia (UTM)

- Ranked among the top-performing students across all 7 semesters of the Bachelor's program, graduating with CGPA 3.79 / 4.00

SKILLS

Technical Proficiency

- **Frontend:** React, JavaScript (ES6+), HTML5, CSS3, Tailwind CSS, RESTful API integration
- **Backend:** Node.js, Express.js, Laravel (PHP), REST API design, JWT auth, rate limiting
- **AI / ML:** Python, PyTorch, Hugging Face Transformers, BERT fine-tuning, RAG pipelines, OpenAI API, Gemini API
- **Databases:** MongoDB, MySQL
- **Cloud & Services:** AWS (EC2, S3, Lambda — via certification), Firebase, Cloudflare R2, Git, GitHub
- **Tools & Practices:** Git, Jira, Agile/Scrum, API Integration, Debugging, Performance Optimization

Professional Skills

- **Problem Solving:** Strong analytical mindset with experience optimizing performance, diagnosing issues, and delivering scalable solutions.
- **Team Collaboration:** Proven ability to work in cross-functional and international Agile teams.
- **Project Ownership:** Experience delivering features end-to-end from design to deployment.
- **Communication:** Clear technical communication with both technical and non-technical stakeholders.

CERTIFICATIONS

AWS Academy Graduate - AWS Academy Cloud Foundations

30 Apr 2025

Amazon Web Services Training and Certification

AWS Academy Graduate - AWS Academy Data Engineering

31 May 2025

Amazon Web Services Training and Certification

AWS Academy Graduate - Machine Learning Foundation

19 Nov 2025

Amazon Web Services Training and Certification

LEADERSHIP AND COMMUNITY

Head of Academic Committee

Nov 2023 – Dec 2024

UTM ISS Egypt 2023/24

- Organized seminars and revision sessions across majors, coordinating speakers and logistics for **200+ attendees**.

Head of Multimedia Committee

Nov 2023 – Dec 2024

UTM ISS Egypt 2023/24

- Led a team of 8 in content creation, photography, and social media — producing **50+ posts** and **event video recaps**

Head of Media Committee

Sep 2024 – Dec 2024

Arabic Cultural Day 2024 – UTM

- Supervised photographers, designers, and writers to execute full media coverage for a university cultural event.

STEM Volunteer Teacher

Feb 2023 – Mar 2023

Amitabha Malaysia Orphanage, AIESEC

Johor Bahru

- Taught Science, English, and Arts to children at an orphanage, developing adaptive instructional strategies.

LANGUAGES

Arabic | English | French